

FastAPI Masterclass

Boris Paskhaver

What is FastAPI?

- A Python framework for building web APIs
- An API is a web server, a continuously running software application on the web that responds to requests.
- **API** is an acronym for “application programming interface”.

History and Popularity

- Developed and maintained by Sebastian Ramirez
- First released in 2018 and still in active development
- Over 100,000 stars on GitHub
- Over 20 million daily downloads on PyPi
- Used by companies including Uber, Netflix, Amazon, Meta, and OpenAI

Features

- Validates data payloads coming into and out of the server
- Parses query parameters, path parameters, cookies, headers, and more
- Optimized for speed (both in terms of performance and developer iteration)
- Generates automatic Swagger/OpenAPI documentation from your code

FastAPI vs. The Competition

- FastAPI has overtaken both Django and Flask in daily downloads.
- FastAPI is not as full-featured as a full framework like Django.
- The developer is responsible for assembling the pieces of the application (database interaction, models, validations, testing, file structure, etc).

Foundations

- FastAPI is built upon two other Python libraries.
- **Starlette** is the underlying asynchronous framework that enables the FastAPI web server.
- **Pydantic** is a data validation library. It both verifies that data fits an expected shape *and* contorts it to fit the expected shape.

Prerequisites

- Intermediate Python
 - Data types, functions, classes, library imports
- Terminal
- Git
- Text editor
 - We'll use Visual Studio Code (VSCode)

What We'll Cover

- FastAPI
- Backend development
- Databases
 - SQL
 - Object Relational Mappers (ORMs)
- Additional Libraries
 - Pydantic
 - SQLAlchemy
- Authentication and authorization

Battle Plan

1 Install the **uv** command-line tool for managing Python projects

3 Work through the **rent-a-room** project

5 Introduce the **FastAPI** full stack template including integration with React frontend

2 Download course materials from GitHub

4 Cover supplementary topics (authentication, testing) in isolation

6 Deploy application to the web